Luxembourg
Luxembourg Competition Authority

Faced with an increasing volume of documents and information, particularly following inspections, the Luxembourg Competition Authority has launched an internal project to strengthen its eDiscovery and data-handling capabilities. The initiative focuses on improving the analysis of seized documents to efficiently identify information relevant to ongoing investigations.

The project is being developed in collaboration with the Luxembourg Institute of Science and Technology (LIST), which brings technical expertise and a suite of adaptable tools. While initially designed for research contexts, these tools are being tailored to meet the specific operational needs of the Authority.

The initiative is structured around three themes.

I. Topic Extraction and Document Clustering

Support document discovery by leveraging topic extraction and organizing documents into thematic clusters, providing investigators with a visual map of the content landscape. This will allow for a more intuitive and efficient exploration of large datasets.

II. Smart Document Management and Search

Improve document annotation, collaboration, and search within those documents. This will be done by integrating keyword and natural language querying via an AI assistant featuring RAG and deep research functionalities.

In collaboration with LIST and the European Commission, a set of AI agents powered by large language models (LLMs) are being developed to assist the Authority with:

- Document classification and labeling (e.g., tagging items based on their respective content)
- Data visualization through customizable dashboards for high-level insights
- Conversational data exploration

III. Social Network Graph Visualization—Mapping Actor Relationships

Extracting and visualizing relationships between entities mentioned in seized documents. The resulting network graphs can reveal connections between individuals, companies, and actions. The first tools are expected to be deployed by the end of the year.